Week 1

Introduction to Operational Research

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Module Handbook

- Let’s start with a look at the module handbook.

- You can find this on the module on Course Resources.

Sensitivity: Internal

Reading and Software



All you need is given in this course.



We will be working with the Pyomo and Simpy Python Libraries. We will also be using Excel.



You can find OR books in the library if you want to widen your reading.

What is Operational Research?

- https://www.theorsociety.com/about-or/



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2025 fees

Member Type	Payment method	Annual Fee
Member	Card / BACS	£105.00
Member	Direct Debit (UK Bank)	£94.00 (or £7.83 p/m)
Concession	Card / BACS	£69.00
Concession	Direct Debit (UK Bank)	£61.50 (or £5.12 p/m)
Student	None	Fully subsidised

OR Society

https://www.theorsociety.com/

Why?

- Most problems don’t have an analytical solution,
- Need models and methods that can provide us with a good solution!
- Finding a good solution to a problem generally has benefits.
- e.g.
  - Monetary
  - Better decision making
  - Time saving
  - Space saving
- Think Covid Modelling as a famous recent example



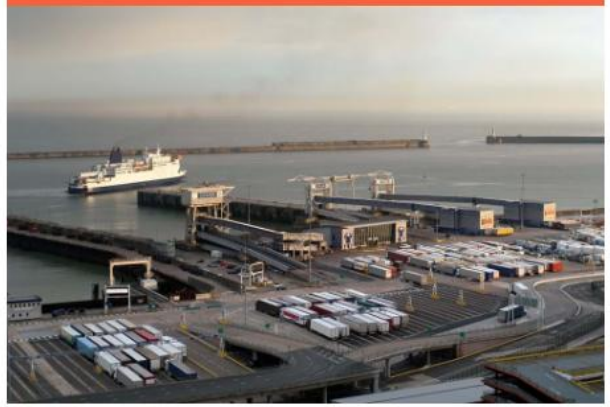
Example fields that use Operational Research

Manufacturing	Production	Mechanics	Engineering
Transportation	Scheduling	Networks	Finance

Just Some Practical Applications…

- Investment - Portfolio Optimization
- Production – Stock Cutting
- Logistics – Distribution, Transportation, Facility Location
- Purchasing – Stock control, i.e. reordering
- Airlines – Crew scheduling
- Sound Engineering – Optimal speaker placement
- Health – Optimising drug regimes, e.g. Cancer Chemotherapy
- Sport – Diet, training plans, tournament scheduling
- Online retail/advertising – Think eBay, Amazon etc…
- To name but a few…

TRAFFIC MODELLING AT THE PORT OF DOVER



THE ART OF CREW SCHEDULING



SCHEDULING HEAT TREATMENTS WITH SOLV



SCHEDULING FAIRNESS IN THE FIFA WORLD CUP



OR in real world situations

[OR in real world situations | ORS]

[Business case studies | ORS]

Task – 10 minutes in pairs

- Pick a business case - Business case studies | ORS

- What is the problem?
- Who are the stakeholders?
- What are the constraints? (resources, time, budget, etc.)
- What are the objectives / performance measures? (e.g. cost, time, quality, customer satisfaction)
- What methods were used?
- What outcomes were achieved (cost savings, time reductions, service improvement, etc.)?

Feedback what you have found out to the class.

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What is this module and what will it do for me?

- This is a general introduction to the field
- It is not a one size fits all module, it takes years to master.
- Even then the best will get it wrong. Think Covid-19.
- It will teach you how to think and reason about a problem.
- It will give you valuable practical skills when applying for jobs.

Career Opportunities

- British Airways (BA) • Home Office • Office for National Statistics (ONS) • IBM • DSTL (Defence and Security Analytics) • RBS • Actica (Management and Technical Consulting) • Amey (Data, analytics and Infrastructure) • BGL (Price Comparison and Insurance Distribution and Outsourcing) • EY (Consulting, Assurance, Tax and Transaction Services) • Jaguar Land Rover • NATS (Air Traffic Control Services) • ROKE (Research, Development and Consultancy in Technology) • TUI

- Statistician • Data scientist • Data analyst • Operational researcher • Operational analyst

[Starting Out in Operational Research | ORS]

Task – 10 minutes in groups of 2/3

- What interests you about Data Science, AI and Maths?
- What sort of career would you like?
- Which industries or problems would you like to work on? (e.g. healthcare, climate change, gaming, finance, transport)
- What skills do you think you'll need to get there? (technical, teamwork, communication, ethics)
- What impact would you like your work to have? (helping people, saving time/money, protecting the planet, creating new tech)
- What excites you most: solving problems, working with people, or building technology?

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History of Operations Research

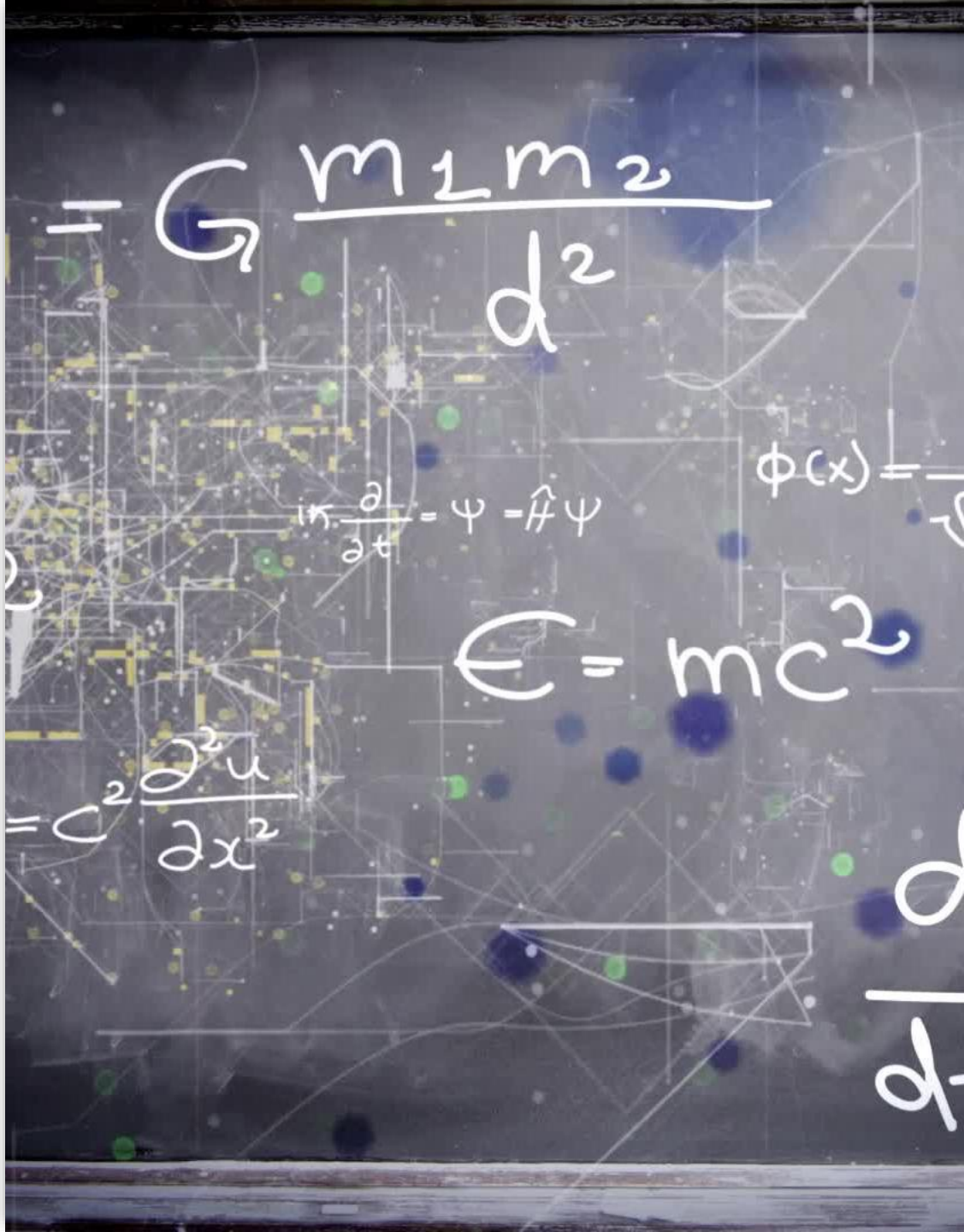
- https://www.youtube.com/watch?v=ILWbaWrjgU4



A brief overview of some Maths

You will cover some of this in other modules, but it will help to get a basic idea of the following to start:

- Linear Equation
- Systems of Equations
- Linear Inequalities
- N-dimensional Space Intuition
  - Hyperplanes
  - Half Spaces
  - Polytope



Linear Equation

This is the basis of Linear Algebra!

1 dimension - Point

$$ax = b$$
$$x \in \mathbb{R}^1$$

2 dimensions - Line

$$ax + by = c$$
$$x \in \mathbb{R}^2$$

3 dimensions - Plane

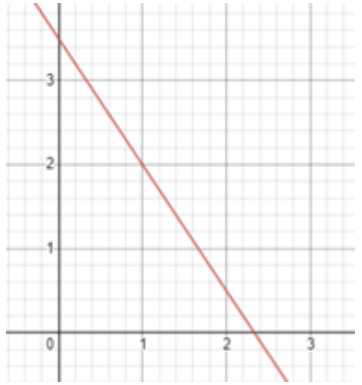
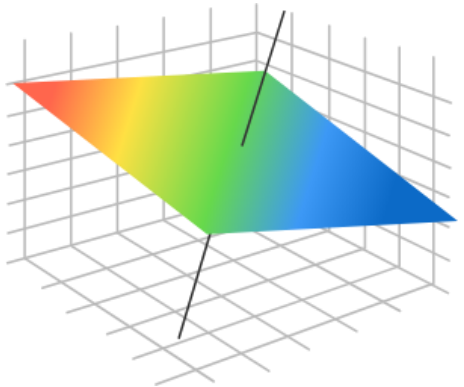
$$ax + by + cz = d$$
$$x \in \mathbb{R}^3$$

>3 dimensions - Hyperplane

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$
$$x \in \mathbb{R}^n$$

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Linear Equation

Dimension	1 dimension	2 dimensions	3 dimensions
Linear Equation	$ax = d$	$ax + by = d$	$ax + by + cz = d$
Visualisation			

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Solving a system of linear equations

$$3x + 2y = 7 \qquad (1)$$
$$2x + 4y = 10 \qquad (2)$$

From (1)

$$x = \frac{7-2y}{3} \qquad (3)$$

Sub (3) into (2)

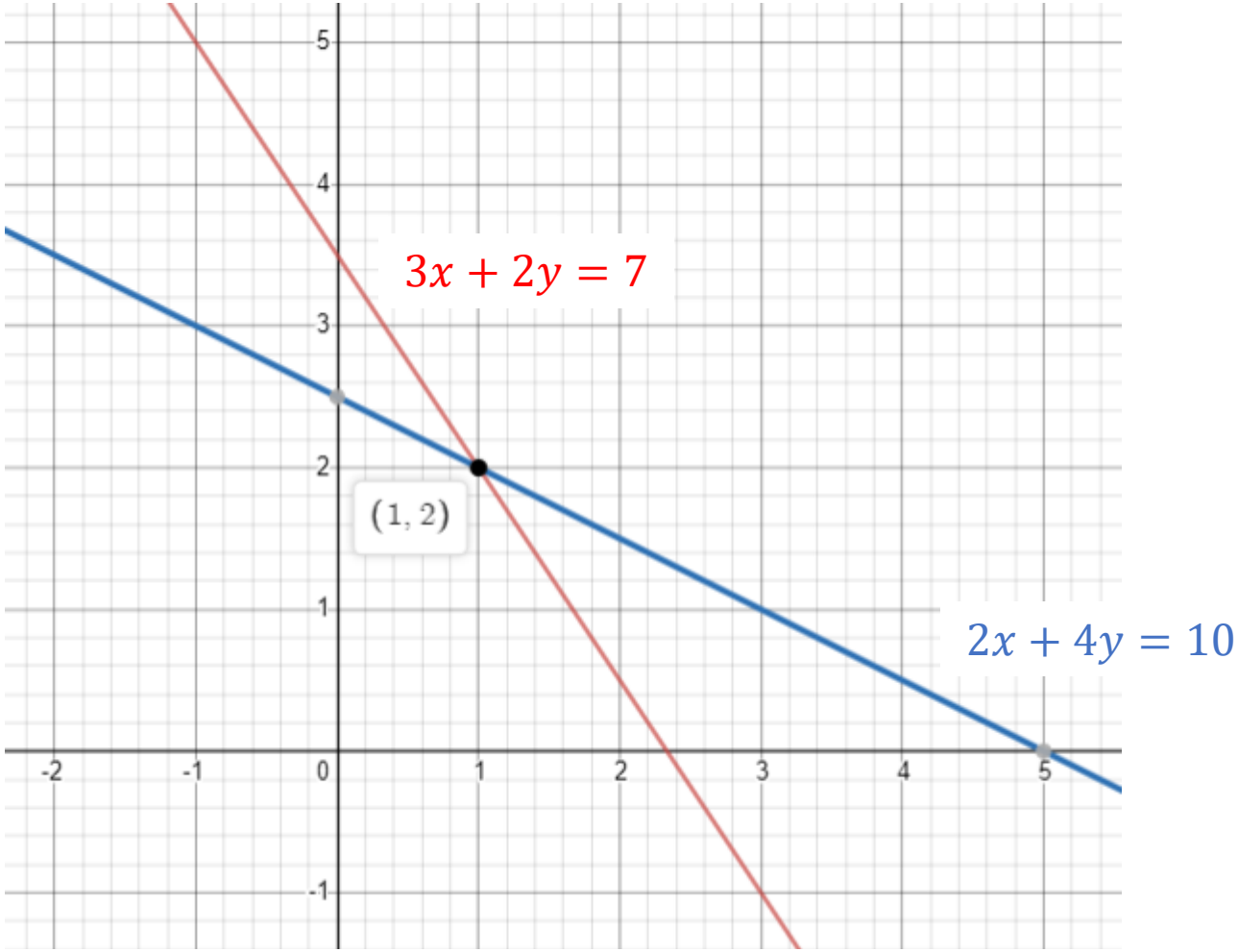
$$2\left(\frac{7-2y}{3}\right) + 4y = 10$$
$$14 - 4y + 12y = 30$$
$$8y = 16$$
$$y = 2$$

From (3)

$$x = \frac{7-4}{3} = 1$$
$$\therefore (x, y) = (1,2)$$

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Row picture



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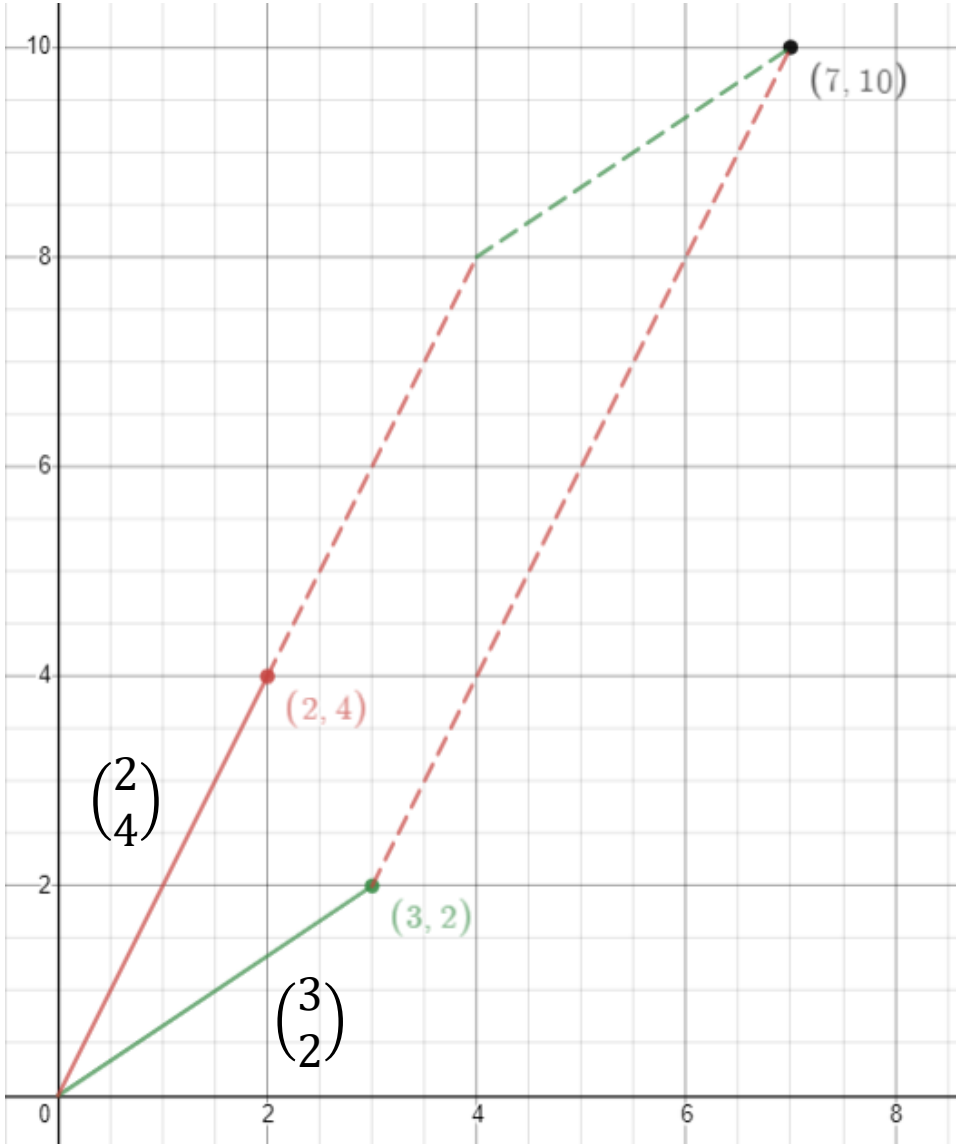
Column picture

$3x + 2y = 7$
$2x + 4y = 10$

We can rewrite this as

$$x\begin{pmatrix}3\\2\end{pmatrix} + y\begin{pmatrix}2\\4\end{pmatrix} = \begin{pmatrix}7\\10\end{pmatrix}$$

Again $x = 1, y = 2$



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Matrix

Inverse of a 2x2 Matrix

$$\boldsymbol{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \boldsymbol{A^{-1}} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

The determinant is defined as follows:

$$|\boldsymbol{A}| = ad - bc$$

$$3x + 2y = 7$$
$$2x + 4y = 10$$

We can rewrite this as

$$\begin{pmatrix} 3 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 10 \end{pmatrix}$$

Solution:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 2 & 4 \end{pmatrix}^{-1}\begin{pmatrix} 7 \\ 10 \end{pmatrix}$$

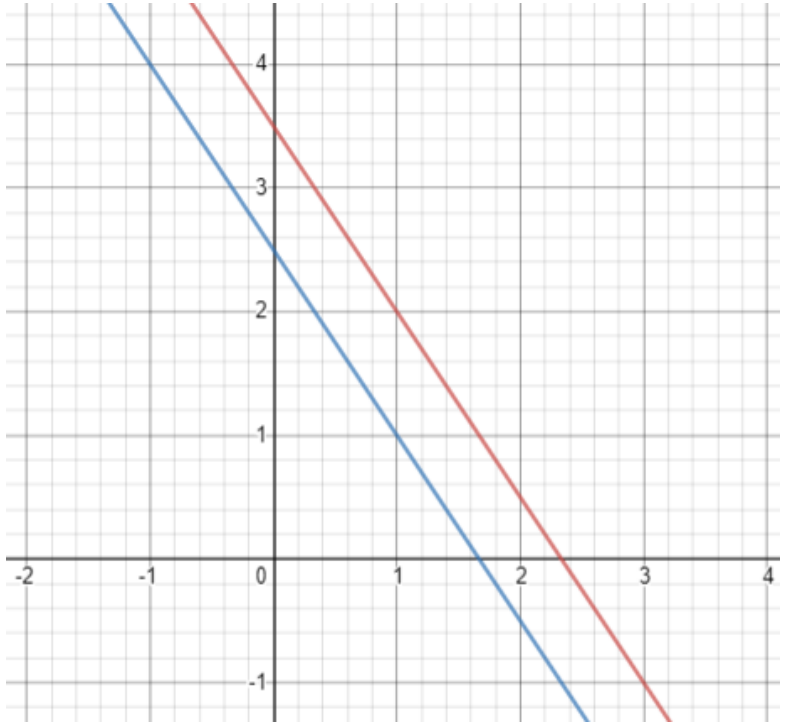
$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{8}\begin{pmatrix} 4 & -2 \\ -2 & 3 \end{pmatrix}\begin{pmatrix} 7 \\ 10 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{8}\begin{pmatrix} 8 \\ 16 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

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What about?

$$3x + 2y = 7$$
$$6x + 4y = 10$$



No solution, lines are parallel.

$$\begin{pmatrix} 3 & 2 \\ 6 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 10 \end{pmatrix}$$

If you try to solve this you get a determinant of 0.

$$3 \cdot 4 - 2 \cdot 6 = 0$$

The matrix is known as singular.

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What about?

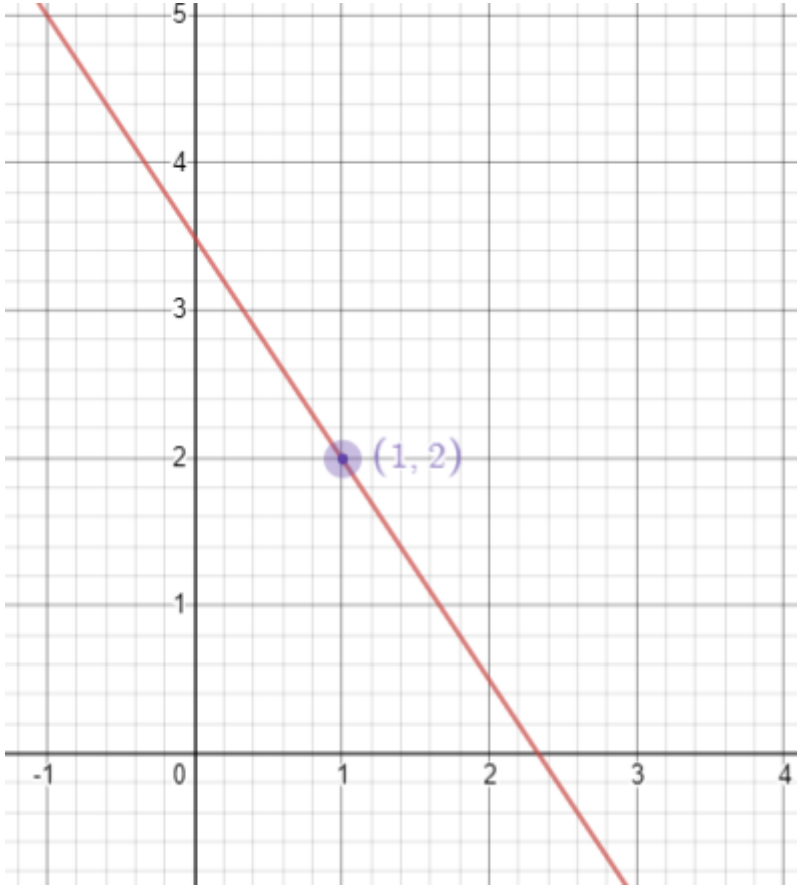
$$3x + 2y = 7$$

More variables than equations

Known as an underdetermined system.

This has infinitely many solutions

$$y = \frac{7}{2} - \frac{3}{2}x$$



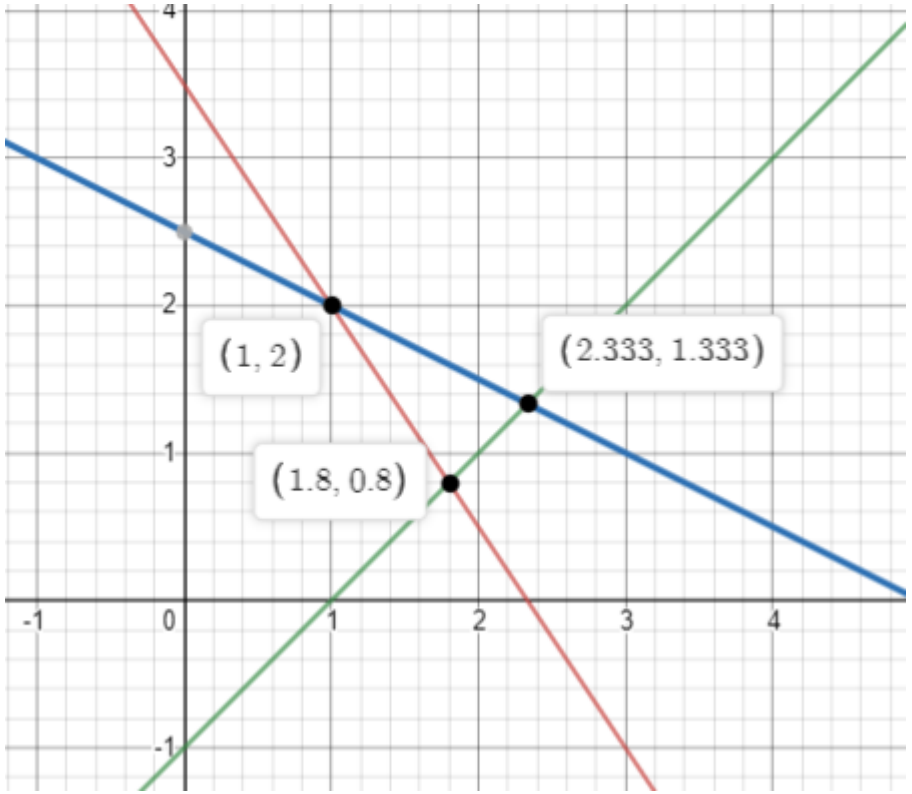
$(x, \frac{7}{2} - \frac{3}{2}x)$ solves the system

In general, has infinitely many solutions, but it may have no solution. e.g 2 parallel planes in $\mathbb{R}^3$

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What about?

$$3x + 2y = 7$$
$$2x + 4y = 10$$
$$x - y = 1$$



More equations than variables

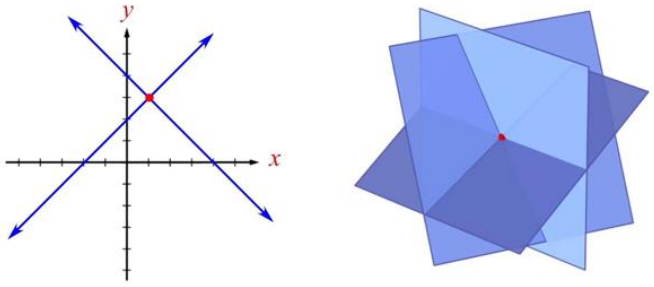
Known as an overdetermined system

Has no solution! i.e. no point solves all three equations simultaneously!

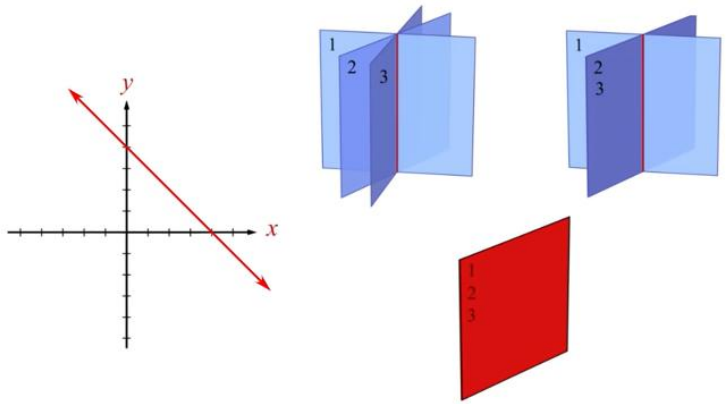
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Visual solutions in 3 dimensions

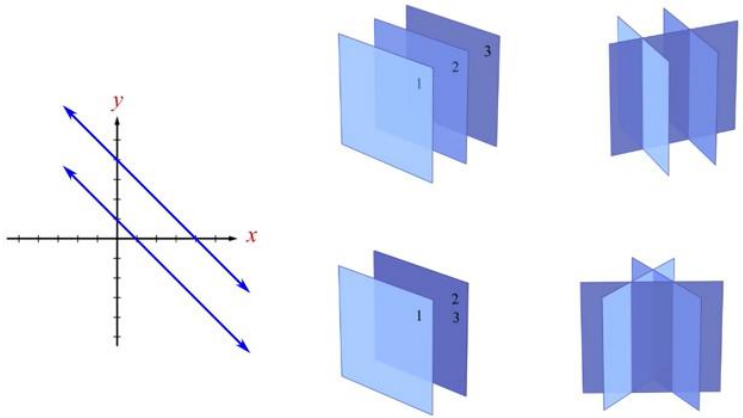




one solution



infinite number of solutions

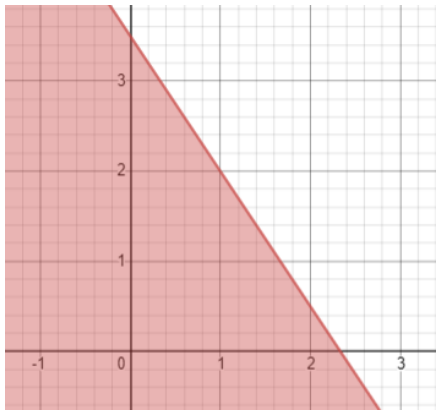
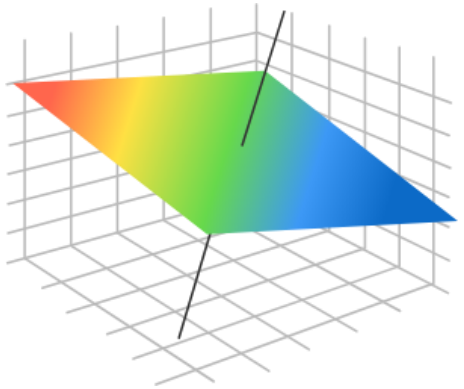


no solutions

https://www.youtube.com/watch?v=WAzUwzV1F3g

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Linear Inequalities

Dimension	1 dimension	2 dimensions	3 dimensions
Linear inequality	$ax \leq d$	$ax + by \leq d$	$ax + by + cz \leq d$
Visualisation			

Sensitivity: Internal

In general

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n \leq b$$

Assume that not all $a_1, \cdots, a_n$ are zero

This is known as a closed half space

Note that we can rewrite this as

$$(a_1 \quad \cdots \quad a_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \leq b$$

$$a \in \mathbb{R}^n, \qquad x \in \mathbb{R}^n$$

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Systems of Linear Inequalities

$$
\begin{aligned}
-x_1 \quad + x_2 &\leq 1 \\
3x_1 + 2x_2 &\leq 12 \\
2x_1 + 3x_2 &\leq 12 \\
x_1 \qquad\quad &\geq 0 \\
x_2 &\geq 0
\end{aligned}
$$

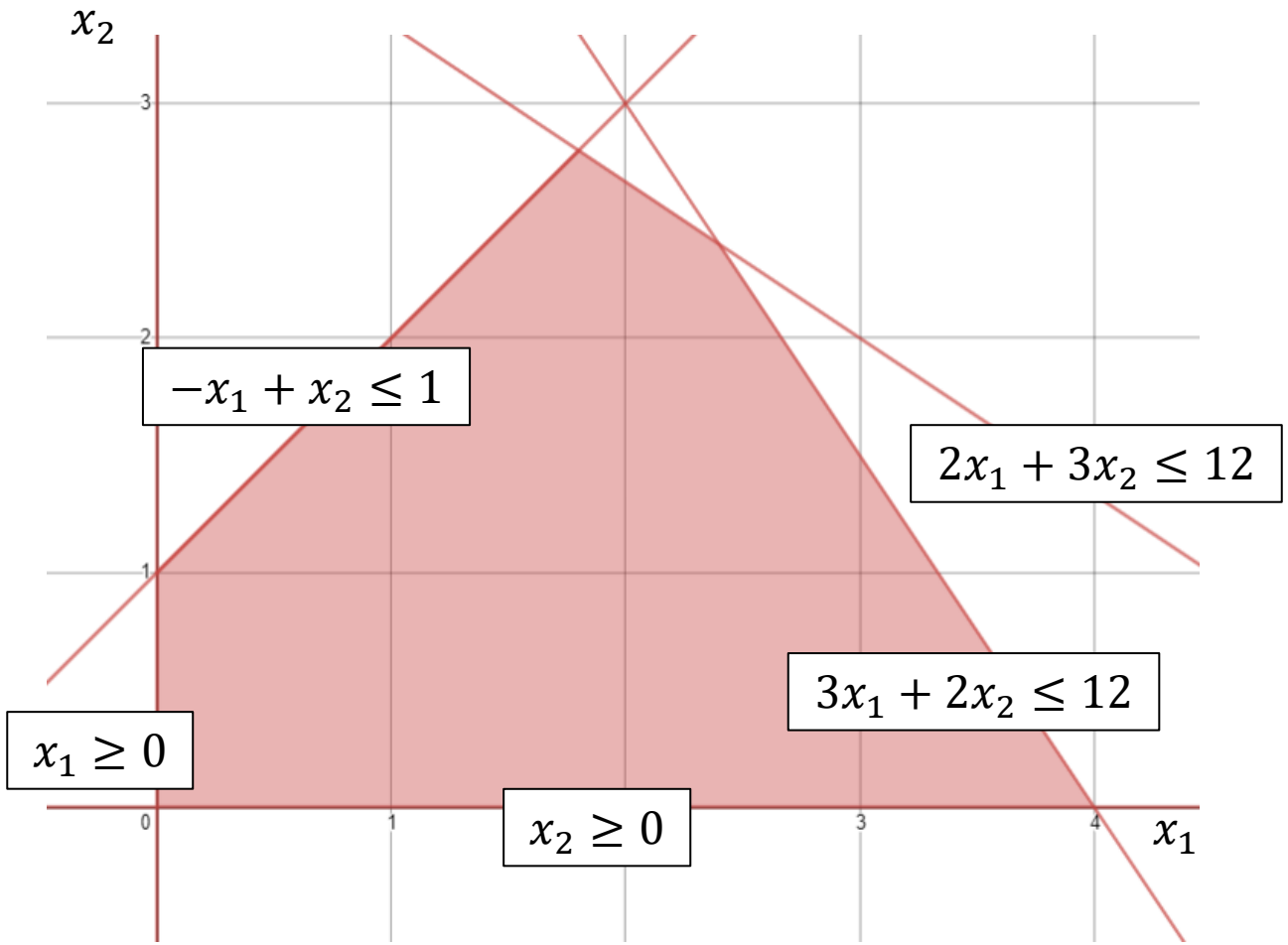
The red area is the **feasible region**.

The half spaces carve out a polygon (red area).

Any point in the feasible region satisfies the system of inequalities.

e.g.

$$(x_1, x_2) = (1,1)$$



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Systems of Linear Inequalities

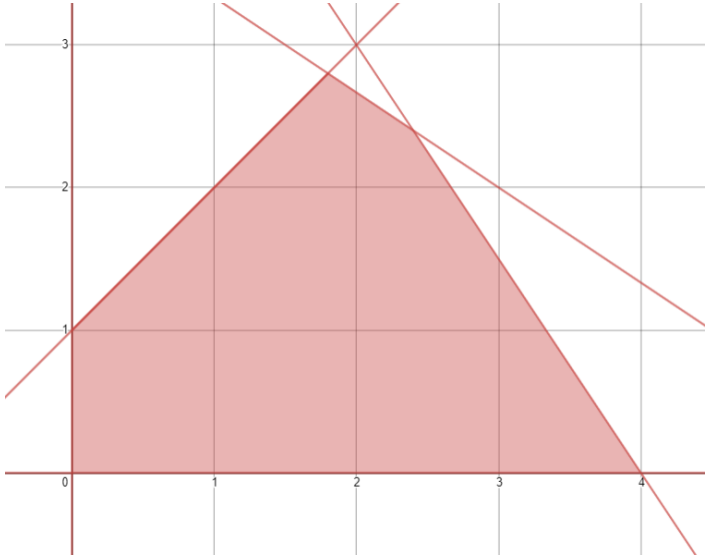
$$\begin{aligned} -x_1 + x_2 &\leq 1 \\ 3x_1 + 2x_2 &\leq 12 \\ 2x_1 + 3x_2 &\leq 12 \\ x_1 &\geq 0 \\ x_2 &\geq 0 \end{aligned} \longrightarrow \begin{aligned} -x_1 + x_2 &\leq 1 \\ 3x_1 + 2x_2 &\leq 12 \\ 2x_1 + 3x_2 &\leq 12 \\ -x_1 &\leq 0 \\ -x_2 &\leq 0 \end{aligned} \longrightarrow \begin{aligned} -1x_1 + 1x_2 &\leq 1 \\ 3x_1 + 2x_2 &\leq 12 \\ 2x_1 + 3x_2 &\leq 12 \\ -1x_1 + 0x_2 &\leq 0 \\ 0x_1 + -1x_2 &\leq 0 \end{aligned}$$

$$\begin{pmatrix} -1 & 1 \\ 3 & 2 \\ 2 & 3 \\ -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 1 \\ 12 \\ 12 \\ 0 \\ 0 \end{pmatrix}$$

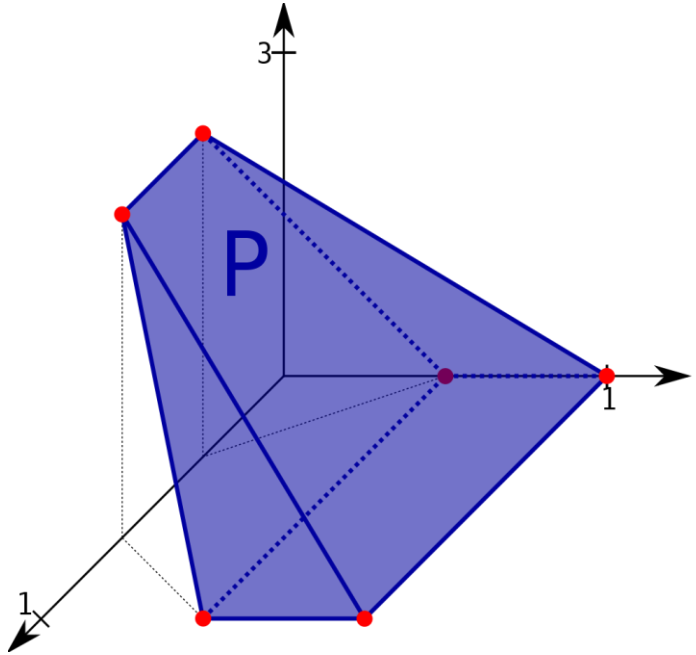
This is the matrix form of the system of inequalities, we will see this a lot!

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Visualisation in 3 dimensions



In 2 dimensions our half-spaces intersect to form a polygon



In 3 dimensions our half-spaces intersect to form a polyhedron

In n dimensions our half-spaces intersect to form a polytope.

Advanced: Both of these shapes are convex

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